Silicon Photonics & Integrated Optics

Module 2.2: Fundamentals of Add-Drop Silicon Microring Resonators

Module Summary

This module expands upon the single-bus architecture by introducing the **add-drop silicon microring resonator**. By coupling the resonant ring to a second bus waveguide, we create a four-port optical device capable of wavelength-selective routing (channel dropping and multiplexing). We rigorously derive the steady-state internal field E_{r1}, the pass-port power transmission T_p (**Eq. 5**), the drop-port power transmission T_d (**Eq. 6**), and the resulting resonance bandwidth FWHM (**Eq. 8**). Particular focus is given to spatial causality, coupler loading effects, and symmetric critical coupling conditions.

1 Physical Intuition & Topology

1.1 The Four-Port Optical Interchange

An add-drop resonator consists of a circular microring placed between two parallel straight bus waveguides:

1. **Input Port:** Injects the multi-wavelength optical signal into the lower bus waveguide.
2. **Pass Port:** Collects light that continues straight through the lower bus without being routed.
3. **Drop Port:** Collects light at resonant wavelengths ($\lambda = \lambda_{\text{res}}$) that couples into the ring and exits through the upper bus in the reverse direction.
4. **Add Port:** Allows an independent signal at λ_{res} to be injected into the upper bus and multiplexed into the forward pass path.

1.2 Why the Internal Field Must Complete a Full Lap to Leak

A critical spatial nuance in understanding microring interference is tracking the position of E_{r1}:

- **Definition of E_{r1}:** E_{r1} is defined as the circulating field **just after Coupler 1**, traveling *away* from the input junction.
- **Spatial Causality:** Light at this exact spatial coordinate cannot immediately leak back out into the pass port because it has already moved past the coupling region.
- **The Recirculation Requirement:** To exit into the pass port, light in E_{r1} must travel around the entire perimeter: passing through the upper arc ($\sqrt{a}e^{j\phi/2}$), traversing Coupler 2 (r_2), completing the lower arc ($\sqrt{a}e^{j\phi/2}$), and finally cross-coupling across Coupler 1 (jk_1). This introduces a full-lap propagation factor $r_2 a e^{j\phi}$.

1.3 Linewidth Broadening: Dual-Coupler Cavity Loading

In an all-pass ring (Module 2.1), light escapes the cavity through only *one* coupler. In an add-drop ring, photons have *two* escape avenues (Coupler 1 and Coupler 2). This additional loss mechanism decreases the photon lifetime (τ_p), lowers the cavity Quality Factor (Q), and significantly **broadens the resonance linewidth** (FWHM).

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2 Key Concepts & Terminology

Add-Drop Configuration A microring coupled to two bus waveguides, functioning as a four-port bandpass/bandstop optical filter.

Coupler 1 (Input Coupler) The directional coupler between the primary bus (Input/Pass) and the microring (r_1, k_1).

Coupler 2 (Drop Coupler) The directional coupler between the secondary bus (Drop/Add) and the microring (r_2, k_2).

Pass-Port Transmission (T_p) The fraction of input power exiting the forward throughput port of the primary bus (notch filter spectrum).

Drop-Port Transmission (T_d) The fraction of input power transferred to the secondary bus (bandpass spectrum).

Half-Lap Phase Accumulation The phase shift ($\phi/2$) and attenuation ($\sqrt{a}$) accrued while traveling halfway around the ring from Coupler 1 to Coupler 2.

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3 Equation Walkthroughs & Derivations

3.1 Part 1: Derivation of the Internal Field (E_{r1})

We track the total steady-state circulating field E_{r1} located immediately downstream of Coupler 1.

1. **Initial Injection:** Light cross-coupling from the input bus into the ring acquires a 90° phase shift (j) and coupling fraction k_1:
$$E_{\text{initial}} = jk_1 E_{\text{input}} \tag{1}$$

2. **First Full Round Trip:** To return to Coupler 1, light travels the full loop length L (loss factor a, phase $e^{j\phi}$), passes straight through Coupler 2 without dropping (r_2), and passes straight through Coupler 1 to remain in the cavity (r_1):
$$E_{\text{lap 1}} = jk_1 E_{\text{input}} \cdot \left(r_1 r_2 a e^{j\phi}\right) \tag{2}$$

3. **Higher-Order Recirculations:** Each subsequent lap adds a factor of $(r_1 r_2 a e^{j\phi})^n$. Summing all circulating contributions:
$$E_{r1} = jk_1 E_{\text{input}} \left[1 + (r_1 r_2 a e^{j\phi}) + (r_1 r_2 a e^{j\phi})^2 + \dots\right] \tag{3}$$

Applying the infinite geometric series sum $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ for $|r_1 r_2 a e^{j\phi}| < 1$:
$$E_{r1} = \frac{jk_1 E_{\text{input}}}{1 - r_1 r_2 a e^{j\phi}} \tag{4}$$

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3.2 Part 2: Derivation of Equation (5) — Pass-Port Transmission (T_p)

The pass-port field E_{pass} is the superposition of the uncoupled direct field and light leaking back out from the ring.

3.2.1 Derivation of Field Transfer Function

1. **Direct Through Field:** $r_1 E_{\text{input}}$
2. **Leakage Field:** Internal field E_{r1} travels a full lap ($r_2 a e^{j\phi}$) and cross-couples out across Coupler 1 (jk_1):

$$E_{\text{leakage}} = E_{r1} \cdot \left(r_2 a e^{j\phi}\right) \cdot (jk_1) = \left(\frac{jk_1 E_{\text{input}}}{1 - r_1 r_2 a e^{j\phi}}\right) \cdot jk_1 r_2 a e^{j\phi} = \frac{-k_1^2 r_2 a e^{j\phi} E_{\text{input}}}{1 - r_1 r_2 a e^{j\phi}} \tag{5}$$

Combining direct pass and leakage terms ($j \cdot j = -1$):

$$\frac{E_{\text{pass}}}{E_{\text{input}}} = r_1 - \frac{k_1^2 r_2 a e^{j\phi}}{1 - r_1 r_2 a e^{j\phi}} = \frac{r_1(1 - r_1 r_2 a e^{j\phi}) - k_1^2 r_2 a e^{j\phi}}{1 - r_1 r_2 a e^{j\phi}} = \frac{r_1 - r_2 a e^{j\phi}(r_1^2 + k_1^2)}{1 - r_1 r_2 a e^{j\phi}} \tag{6}$$

For a lossless coupler, conservation of power dictates $r_1^2 + k_1^2 = 1$:

$$\frac{E_{\text{pass}}}{E_{\text{input}}} = \frac{r_1 - r_2 a e^{j\phi}}{1 - r_1 r_2 a e^{j\phi}} \tag{7}$$

3.2.2 Derivation of Power Transmission (T_p)

Multiplying Eq. (7) by its complex conjugate:

$$T_p = \left|\frac{E_{\text{pass}}}{E_{\text{input}}}\right|^2 = \frac{(r_1 - r_2 a e^{j\phi})(r_1 - r_2 a e^{-j\phi})}{(1 - r_1 r_2 a e^{j\phi})(1 - r_1 r_2 a e^{-j\phi})} \tag{8}$$

Expanding using Euler's identity ($e^{j\phi} + e^{-j\phi} = 2\cos\phi$):

$$\text{Numerator} = r_1^2 - r_1 r_2 a(e^{j\phi} + e^{-j\phi}) + r_2^2 a^2 = r_1^2 - 2r_1 r_2 a \cos\phi + r_2^2 a^2 \tag{9}$$

$$\text{Denominator} = 1 - r_1 r_2 a(e^{j\phi} + e^{-j\phi}) + (r_1 r_2 a)^2 = 1 - 2r_1 r_2 a \cos\phi + (r_1 r_2 a)^2 \tag{10}$$

Rearranging terms yields **Equation (5)**:

$$T_p = \frac{r_2^2 a^2 - 2r_1 r_2 a \cos\phi + r_1^2}{1 - 2r_1 r_2 a \cos\phi + (r_1 r_2 a)^2} \tag{11}$$

3.2.3 Numeric Sanity Check

Given $r_1 = r_2 = 0.9$ and $a = 0.9985$ ($r_1 r_2 a \approx 0.8088$):

- **On-Resonance ($\cos\phi = 1$):**

$$T_p = \frac{(r_2 a - r_1)^2}{(1 - r_1 r_2 a)^2} = \frac{(0.9 \times 0.9985 - 0.9)^2}{(1 - 0.8088)^2} = \frac{(-0.00135)^2}{(0.1912)^2} \approx 4.99 \times 10^{-5} \quad (-43\,\text{dB notch depth}) \tag{12}$$

- **Off-Resonance ($\cos\phi = -1$):**

$$T_p = \frac{(r_2 a + r_1)^2}{(1 + r_1 r_2 a)^2} = \frac{(0.89865 + 0.9)^2}{(1 + 0.8088)^2} = \frac{(1.79865)^2}{(1.8088)^2} \approx 0.9888 \quad (98.9\%\ \text{power passed}) \tag{13}$$

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3.3 Part 3: Derivation of Equation (6) — Drop-Port Transmission (T_d)

To find E_{drop}, we track light from E_{r1} as it travels a half-lap to Coupler 2 and cross-couples into the drop waveguide.

3.3.1 Derivation of Field Transfer Function

1. **Half-Lap Propagation:** Accumulates half the loss ($\sqrt{a}$) and half the phase ($e^{j\phi/2}$):

$$E_{\text{at Coupler 2}} = E_{r1} \cdot \sqrt{a} e^{j\phi/2} \tag{14}$$

2. **Cross-Coupling into Drop Bus:** Acquires phase shift j and coupling fraction k_2:

$$E_{\text{drop}} = E_{\text{at Coupler 2}} \cdot (jk_2) = (E_{r1}\sqrt{a}e^{j\phi/2}) \cdot (jk_2) \tag{15}$$

Substituting Eq. (4) into E_{drop} ($j \cdot j = -1$):

$$\frac{E_{\text{drop}}}{E_{\text{input}}} = \frac{-k_1 k_2 \sqrt{a} e^{j\phi/2}}{1 - r_1 r_2 a e^{j\phi}} \tag{16}$$

3.3.2 Derivation of Power Transmission (T_d)

Multiplying Eq. (16) by its complex conjugate:

$$T_d = \left| \frac{E_{\text{drop}}}{E_{\text{input}}} \right|^2 = \frac{(-k_1 k_2 \sqrt{a} e^{j\phi/2})(-k_1 k_2 \sqrt{a} e^{-j\phi/2})}{(1 - r_1 r_2 a e^{j\phi})(1 - r_1 r_2 a e^{-j\phi})} \tag{17}$$

Note that $e^{j\phi/2} \cdot e^{-j\phi/2} = 1$ and $\sqrt{a} \cdot \sqrt{a} = a$:

$$\text{Numerator} = k_1^2 k_2^2 a \tag{18}$$

Substituting power conservation relations $k_1^2 = 1 - r_1^2$ and $k_2^2 = 1 - r_2^2$ yields **Equation (6)**:

$$T_d = \frac{(1 - r_1^2)(1 - r_2^2)a}{1 - 2r_1 r_2 a \cos\phi + (r_1 r_2 a)^2} \tag{19}$$

3.3.3 Numeric Sanity Check

Given $r_1 = r_2 = 0.9$, $k_1^2 = k_2^2 = 0.19$, and $a = 0.9985$:

- **On-Resonance ($\cos\phi = 1$):**

$$T_d = \frac{(1 - 0.81)(1 - 0.81)(0.9985)}{(1 - 0.8088)^2} = \frac{(0.19)(0.19)(0.9985)}{(0.1912)^2} = \frac{0.03604}{0.03656} \approx 0.9858 \quad (98.6\% \text{ power dropped}) \tag{20}$$

- **Off-Resonance ($\cos\phi = -1$):**

$$T_d = \frac{0.03604}{(1 + 0.8088)^2} = \frac{0.03604}{3.2717} \approx 0.0110 \quad (1.1\% \text{ power dropped}) \tag{21}$$

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3.4 Part 4: Equation (8) — Resonance Bandwidth (FWHM)

$$\text{FWHM} = \frac{(1 - r_1 r_2 a)\lambda_{\text{res}}^2}{\pi n_g L \sqrt{r_1 r_2 a}} \tag{22}$$

3.4.1 Plain Words

Equation (22) calculates the full width at half maximum spectral bandwidth of an add-drop ring filter.

3.4.2 Symbol Definitions

Symbol	Units	Physical Description
FWHM	m (or nm)	Spectral resonance width at 3 dB bandwidth points
r_1, r_2	–	Self-coupling field coefficients for Couplers 1 and 2
a	–	Round-trip amplitude attenuation factor
$\lambda_{\mathbf{res}}$	m	Resonant wavelength
n_g	–	Group index ($n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}$)
L	m	Perimeter round-trip length ($L = 2\pi R$)

3.4.3 Derivation & Physical Interpretation

The derivation follows the standard Lorentzian cavity width evaluation, replacing the single-coupler reflection r of the all-pass ring with the combined two-coupler effective reflection $r_1 r_2$.

3.4.4 Numeric Comparison: All-Pass vs. Add-Drop Bandwidth

Using consistent waveguide parameters ($\lambda_{\text{res}} = 1.55\,\mu$m, $L = 62.83\,\mu$m, $n_g = 4.2$, $a = 0.9985$):

- **All-Pass Ring ($r = 0.9$):**

$$\text{FWHM}_{\text{all-pass}} = \frac{(1 - 0.9 \times 0.9985)(1.55 \times 10^{-6})^2}{\pi \times 4.2 \times (6.283 \times 10^{-5})\sqrt{0.89865}} \approx 0.153\,\text{nm} \tag{23}$$

- **Add-Drop Ring ($r_1 = r_2 = 0.9$):**

$$\text{FWHM}_{\text{add-drop}} = \frac{(1 - 0.8088)(1.55 \times 10^{-6})^2}{\pi \times 4.2 \times (6.283 \times 10^{-5})\sqrt{0.8088}} \approx 0.616\,\text{nm} \tag{24}$$

Key Physical Takeaway

The add-drop filter bandwidth (0.616 nm) is **4 times broader** than the all-pass filter bandwidth (0.153 nm). The addition of the second coupler creates a second extraction pathway for circulating photons, lowering cavity lifetime and widening the filter passband.

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4 Critical Coupling Conditions in Add-Drop Rings

To achieve complete extinction ($T_p = 0$) at the pass port on resonance ($\cos\phi = 1$), the numerator of Eq. (11) must equal zero:

$$(r_2 a - r_1)^2 = 0 \implies r_1 = r_2 a \tag{25}$$

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Coupling Regime	Condition	Operational Consequences
Symmetric / Balanced	$r_1 = r_2$	If $a \approx 1$, $r_1 \approx r_2 a$. Achieves near-zero pass transmission ($T_p \approx 0$) and maximum drop efficiency ($T_d \approx 1$).
Under-Coupled	$r_1 > r_2 a$	Input coupling is weaker than combined internal and drop-port losses. Incomplete pass extinction.
Over-Coupled	$r_1 < r_2 a$	Input coupling dominates. Broadened drop bandpass, degraded pass notch depth.

5 Check Your Understanding

Check Your Understanding

Question 1: An add-drop microring filter has identical input and drop coupling coefficients ($r_1 = r_2 = 0.92$) and a round-trip loss of $a = 0.95$. Is the pass port critically coupled?

Answer: No. Critical coupling requires $r_1 = r_2 a$. Here, $r_2 a = 0.92 \times 0.95 = 0.874 \neq 0.92$. Since $r_1 > r_2 a$, the device is **under-coupled**.

Question 2: What is the peak drop-port efficiency (T_d) of a completely lossless ($a = 1.0$) symmetric ($r_1 = r_2 = r$) add-drop ring at resonance?

Answer: Setting $a = 1$ and $r_1 = r_2 = r$ in Eq. (19) at resonance ($\cos\phi = 1$):

$$T_d = \frac{(1 - r^2)^2}{(1 - r^2)^2} = 1.0 \quad (100\% \text{ efficiency}) \tag{26}$$

All optical energy is routed seamlessly from the input port to the drop port without reflection back into the pass port.